

Technology Ties:

The Rise and Roles of Military AI Strategic Partnerships

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(working paper: feedback welcome)

Abstract: Emerging and Disruptive Technologies (EDTs) are reshaping military institutions and challenging traditional understandings of these actors' tools, environment, and mission. This article explores how the proliferation of artificial intelligence (AI) technologies is set to drive a new generation of strategic partnerships, focused on nurturing and shaping that technology's military application. These 'military AI strategic partnerships' are distinct from both traditional alliances and from strategic partnerships for other technologies, and may have significant implications for AI's future military use, as well as for geopolitics. To investigate this phenomenon, this article explores the concept of a strategic technology partnership, discussing the uses, practices, and distinct operational requirements involved in military AI strategic partnerships. It then examines four recent cases: (1) the US-led AI Partnership for Defense (PfD); (2) the AUKUS partnership; (3) robust China-Russia cooperation on military AI; and (4) transatlantic (especially US-UK) cooperation. The article discusses the implications of these partnerships for broader AI governance stakeholders, standing military alliances like NATO, and for the development and usage of military AI itself.

Keywords: artificial intelligence, military AI, Emerging and Disruptive Technologies, strategic partnerships, military AI strategic partnerships, Partnership for Defense, AUKUS, China-Russia cooperation.

Introduction

Technological change is frequently an exercise in political realignment and recalibration, particularly at the international level. These changes, Daniel Drezner noted, can “create new winners and losers, alter actor preferences, and allow the strategic construction of new norms and organizations” (Drezner 2019, 287). Throughout history, technological innovations often have had significant political impacts, supporting the early rise of states

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(Turchin et al. 2021), and the staggered development of international law amongst them (Picker 2001). Innovation also frequently has geopolitical implications, contributing to arms races and technological superiority, and rebalancing alliances and partnerships.

The world seems set for such a moment of realignment. Recent years have seen considerable progress in a range of so-called Emerging and Disruptive Technologies (EDTs) (B. Roberts 2021), from quantum technology to biotechnology, and from hypersonic or directed energy weapons, to artificial intelligence (AI) (Congressional Research Service 2021b). This same period has seen increasingly militarized flare-ups of long-running conflicts, both between states — such as the war between Russia and Ukraine — and non-state actors — such as the October 2023 hostilities between Israel and Hamas. It is generally held that EDTs will reshape—indeed are already reshaping—military institutions and the conduct of such conflicts. In doing so, they will challenge traditional understandings of international actors’ tools, environment, and goals. As technological innovation accelerates and expands tactical and strategic capabilities, and as the world is rocked by significant new wars from Europe to the Middle East, scholars of the international security environment must address and account for these changes.

Of this broad portfolio of EDTs, perhaps none has been subject to as much anticipation as AI. The last decade, and especially the last few years, have seen tremendous developments in AI technology, resulting in increasingly widespread uptake in diverse industries and domains of life (Clark and Perrault 2023; Benaich and Hogarth 2022). The rapid innovation in ‘generative AI systems’, such as OpenAI’s ChatGPT or Anthropic’s Claude, and the rise of military applications of large language models in systems like Palantir’s ‘AIP for Defense’ (Palantir 2023), there has been a reignited debate over the role and relevance of AI as a transformative strategic capability, with significant impacts on both civilian and military domains.

It is not surprising that most discussion of the intersection of military AI with geopolitics focuses on *conflictive* relations. As AI is increasingly perceived as a key strategic asset, it has certainly driven significant geopolitical contestation, with constant allegations of a race for supremacy in AI technology (Auslin 2018; But see Roff 2019; Kania 2018), as well as, *inter alia*, competition over access to semiconductor supply chains (F.P. Analytics 2021; S. M. Khan 2021; S. Khan 2019), as well as over AI talent (Huang and Arnold 2020). Recently, the Biden administration announced a ban on chip exports to China out of a growing concern of technology innovation in the Chinese military (Freifeld 2023). The step has significant economic impact and marks “a historic shift in US-China tech policy” (ibid).

Such developments continue to cause friction between the United States (US) and China on military technology development (Johnson 2019).

Strategically powerful technologies can induce not just geopolitical conflict but also (calculated) collaboration. Artificial intelligence is clearly changing traditional security structures and institutions. In that sense, it is the most prominent amongst a suite of EDTs which have begun to shift our understanding of the nature and character of modern warfare, even as they challenge and erode extant weapon technology regimes and security governance arrangements (Nelson 2019; Brockmann 2019). In response, states have been adapting their developmental strategies. As such, the past few years have seen the emergence of new patterns of state collaboration around military AI, often outside pre-existing or legacy institutions. In this article, we provide a snapshot of four cases in particular. We examine the establishment of the AI Partnership for Defense (PfD) in September 2020; the establishment of the AUKUS security pact between the US, UK and Australia a year later; as well as ongoing collaborative initiatives between China and Russia, and trans-Atlantic cooperation, particularly between the US and UK.

By and large, these developments have remained relatively underexplored in recent scholarship on AI, and the strategic impacts of international cooperation. However, recent conflicts illustrate the urgency of examining military technology partnerships; for example, the reported use of weapon systems with autonomous functions from both Russia and Ukraine in the ongoing conflict (Kahn 2022). As the technology landscape is shifting, the time is ripe to map and examine matched changes in the security environment.

This article therefore aims to investigate the role of strategic partnerships around the development and use of military AI technology in order to explore the changing landscape of state collaboration on technology. For each of the four partnerships discussed, we identify the actual cooperation mechanisms members use to operationalize cooperation. As such, we aim to answer the following questions: how do these strategic partnerships for military AI matter? Do they differ from previous forms of international cooperation or partnerships, and if so, in what ways? How do these partnerships facilitate technology cooperation? And what are the implications for geopolitics, and for AI itself? In reviewing these cases, this article will argue that new technology strategic partnerships are changing the landscape of technological development and alliance politics. As technological innovation and superiority become critical facets of military power and capability, it is vital to examine how these—and future—military AI strategic partnerships will influence the geopolitical environment or disrupt the international security landscape.

It is important to recognize that this is an early examination of a fast-developing technological and geopolitical domain. It is therefore aimed at mapping a dynamic landscape and charting the importance and direction of these developments. Nonetheless, there is a premium on having these discussions today, because doing so might allow us to seize opportunities in creating more productive or effective partnerships in this space. This can have path-dependent effects not just on the responsible development of military AI, but also the AI governance ecology more broadly.

In section one, we first review the background of developments in military AI, and discuss the work aimed at mapping international cooperation around it, providing a rationale for examining strategic partnerships. In section two, we set the conceptual foundations for this debate, by discussing the concept of strategic partnership, and the particular challenges facing military AI strategic partnerships. In section three, we then turn to the four cases and sketch developments in this landscape. In section four, we provide a preliminary analysis and review of main themes. Finally, we conclude, and sketch future areas of research.

1. Background & Rationale: Developments in Military AI

Why study strategic partnerships around military AI? Given how recent these developments are, it might be no surprise that the role of strategic partnerships has been somewhat under-examined within mainstream scholarship on AI's international impact. For instance, in the field of AI governance, much of the focus has been on domestic governance and regulation;³ likewise, studies of AI governance at the international level have explored at length the role of various tools or approaches.⁴

Yet many of these governance initiatives and levers explicitly or implicitly exclude discussion of AI's military uses. What then will determine or shape AI's military practices? To be sure, this question is hardly neglected within scholarship or policy. However, within research on military AI, there has been a focus on lethal autonomous weapons systems (LAWS), their (im)possible compliance with international humanitarian law (IHL)

³ Such work has explored, for instance, how AI's governance might be shaped by different state institutions (Scherer 2016; Schiff 2022), different actors in the AI development landscape (Leung 2019; Belfield 2020), or different regulatory tools ranging from corporate governance (Cihon, Schuett, and Baum 2021), to forms of soft law (Gutierrez and Marchant 2021), or hard regulation (Nemitz 2018; M. M. Maas 2022).

⁴ This includes, *inter alia*, discussions of the constellation of existing public international law (Kunz and Ó hÉigeartaigh 2021; Vöneky 2020); ethical principles and frameworks for AI (Fjeld et al. 2019; Stix 2021); the role of standard-setting bodies (Cihon 2019; Lorenz 2020); or science diplomacy (Feijóo et al. 2020). There has also been a focus on the emerging yet incipient regime architecture of multilateral cooperation around AI governance (Schmitt 2022; Cihon, Maas, and Kemp 2020; M. M. Maas and Villalobos 2023), as well as the regulation of AI under the upcoming EU AI Act (Veale and Borgesius 2021; Schuett 2023). For a general review of much of this research, see also (M. Maas 2023)

(Anderson, Reisner, and Waxman 2014; Trabucco and Heller 2022), and their attempted regulation within various frameworks, norms or regimes (Chavannes, Klonowska, and Sweijs 2020; M. M. Maas 2019; Rosert and Sauer 2020), including most prominently within the UN Convention on Certain Conventional Weapons (UN CCW) (Sauer 2021; UNODA n.d.). However, this has not yet resulted in any multilateral solutions, with the December 2021 Sixth Review Conference of the CCW failing to launch negotiations into an international treaty (Farge 2021). Nonetheless, in March 2023, the US, UK, Japan, Australia, and South Korea proposed draft articles toward clarifying the legal requirements for deploying LAWS and specifies measures to reach those requirements (Group of Governmental Experts on Emerging Technologies in the Area of Lethal Autonomous Weapons System 2023).

Despite these developments, and while there has been significant treatment of AI's broader geopolitical implications (Horowitz 2018), there has been less focus on *multilateral efforts to cooperate and collaborate on military AI development* and to examine the broader implications of these strategic partnerships. There is also some attention to prospects for transatlantic cooperation between the EU and US on AI (Newman 2021), as well as its friction points (H. Roberts et al. 2021; Can and Kaplan 2020). Yet again, this has generally not focused closely on pathways of cooperation for the development or deployment of AI systems in *military* contexts specifically. Other existing work does explore military AI, but has focused on the internal development practices of individual states (Bode et al. 2023); or explored or encouraged US cooperation with allies on the 'democratic way of AI' (Imbrie et al. 2020). There is limited attention to bilateral arrangements, barring some calls for confidence-building measures between states around military AI (Horowitz and Scharre 2021). The most comprehensive review in this space is Zoe Stanley-Lockman's exploration of the 'Military AI Cooperation Toolbox' (Stanley-Lockman 2021). Notwithstanding these incisive mappings, there remains somewhat of a gap in attention to AI strategic partnerships, and the greater implications of them. Critically, however, recent developments suggest that such AI strategic partnerships are beginning to accelerate. Indeed, while they take varied forms and involve distinct cooperation mechanisms, they may be in a position to take a large role in shaping military AI development and potentially shaping a significant share of military practices around AI, especially if multilateral and normative legal instruments fail to reach implementation.

Strategic partnerships will therefore likely matter both to the shape of defense collaboration in AI, as well as to how well actors can anticipate, and mitigate challenges around AI safety, interoperability, and strategic instability (Rudner and Toner 2021a). Indeed,

the value of working with partners in this space plausibly offers an asymmetric asset in maintaining a technological edge in the era of strategic competition (Stanley-Lockman 2021). As Zoe Stanley-Lockman notes, “...there is a consensus that strengthening alliances and partnerships is important [...] because the United States acts in coalitions” with vital military partners (Ibid, pg. 1). These efforts by the US and other actors to facilitate and promote military AI cooperation will have significant stakes in the geopolitical context. It will have implications not just for what AI systems are used, or how, but also what will be their risks—and what kind of interventions or policies can help mitigate these risks.

2. The concept of a Strategic Partnership

States have always had mixed incentives regarding collaboration and cooperation on emerging technologies and their military applications. Intuitively, one would expect states to limit information sharing and keep key technological information and tacit knowledge constrained to maintain an advantage over strategic competitors. Nevertheless, there are unique strategic benefits to partnering with allies to cooperate on new technology through resource multiplication, enhanced interoperability in joint operations, and strengthening military partnerships. But firstly, we need to define strategic partnerships and understand why these forms of cooperation have been on the rise.

A. From Alliances to Strategic Partnerships

Basrur and Kutty have offered one definition of strategic partnerships as “cooperative relationships that carry features of strategic (and other forms of) cooperation, but stop short of commitments to fight on behalf of a partner” (Basrur and Kutty 2018, 6). These stand in contrast to traditional alliances (such as NATO) which entail certain security assurances and guarantees as a core part of their *raison d’être*. However, this remains a relatively loose and negative definition of strategic partnerships—which mostly still defines them by what they are not. As such, Basrur and Kutty expand their definition, by noting that strategic partnerships have in common a number of features, namely:

- “Unlike alliances, [strategic partnerships] are **not expressly aimed at an identified enemy**. This leaves space for building bridges with strategic adversaries with whom conflict is not a viable option owing to interdependence.
- Also unlike alliances, strategic partnerships are **limited and flexible and do not involve deep action-inducing commitments to the security or even political positions of partners**.

Typically, states do not take strongly supportive positions on the disputes of strategic partners with other states.

- Strategic partnerships involve **regular high-level political interactions**, including meetings between heads of government and civilian and military officials at lower rungs.
- They **frequently involve security cooperation** by means of arms transfers, joint military exercises, strategic dialogues and intelligence exchanges.
- Beyond security, strategic partnerships usually **extend to other mutually beneficial interactions**, especially economic exchanges.” (Basrur and Kutty 2018, 7, emphasis ours)

We largely agree with Basrur and Kutty’s sketch of strategic partnerships, but offer one caveat that is evident in the case of AI, or even broader technological, strategic partnerships. Strategic partnerships can exist in various forms - some consisting of formal agreements with political commitments, and others resembling loose forms of technological cooperation. Additionally, Basrur and Kutty do not go so far as to identify actual mechanisms to operationalize strategic partnerships. We argue the four case studies detailed below demonstrate that strategic partnerships exist along a spectrum with different designs, arrangements, and degrees of commitment. Nevertheless, each serves to signal a strategic political alignment using technological cooperation as a basis for future security cooperation.⁵

What is the significance of this distinct category of cooperation? Basrur and Kutty argue that there has been a historical shift away from traditional alliances towards strategic partnerships, because of the rise in global interdependence and multipolarity (Basrur and Kutty 2018, 6–7). But why do we see this shift? What do states gain from such arrangements in the absence of military support? Basrur and Kutty argue that by entering into a strategic partnership, states gain a range of benefits such as augmenting their military capabilities; applying joint psychological- political pressure to adversaries; obtaining political support in international institutions, in terms of membership access or support on issues or initiatives, among others. Importantly, by pursuing these interests through a strategic partnership rather than an alliance, they benefit because these goals can be achieved “without entangling the state in an alliance that may drag it into a confrontation with its partner’s adversaries” (Ibid, 8). Moreover, states are able to spread their bets through multiple partnerships, thereby

⁵ It is necessary to acknowledge upfront that strategic partnerships can be either formal commitments or loose partnerships because it may be easy to interpret these partnerships as fundamentally different in nature. Yet while there are important differences, these do not preclude useful analysis and comparison, given that, at the core, each case detailed below ultimately represents a strategic partnership that advances technological advantage through cooperation and collaboration. These types of partnerships, whether formal or informal, will prioritize and advance AI as a critical military capability.

avoiding full dependence on certain powerful partners. Finally, a state might want to pursue a strategic partnership even with an adversary, in order to “tie [that adversary] into a relationship of engagement and cooperation, in order to alleviate risks associated with conflicts of interest” (Ibid).

Notably, it is not enough to simply identify such partnerships. We aim to also identify the mechanisms these partnerships employ to facilitate technology cooperation amongst partners and compare the depth and breadth of the cooperation mechanisms across partnerships. We will be able to see how partnerships embed distinct cooperation practices and draw conclusions about the opportunities, and constraints, of each strategic partnership.

B. Strategic Partnerships for Technology & Military AI

In principle, a strategic partnership can cover many distinct goods, from economic exchanges to security cooperation in the form of arms transfers or exercises. Within the broad category of strategic partnerships, however, there is a subset that are of particular interest in our context. These are strategic *technology* partnerships: forms of strategic partnerships that focus in particular on the shared and/or mutually facilitated development of certain EDTs. Some of these partnerships are focused on the development of industrially or strategically valuable non-military technology (e.g. 5G); others, however, focus on military technology. Strategic technology partnerships focused on the accelerated development of military (or strongly dual-use) technologies have included collaborations aimed at a diverse set of technologies, from fighter aircraft (Birkler, Lorell, and Rich 1997) to hypersonic missiles (Congressional Research Service 2021a, 8), to drones (Mayer 2021) quantum computing (The White House 2021), or cyber (Renard 2018), to name just a few.

A key consideration is how strategic technology partnerships are different from other strategic partnerships, in terms of what is shared, or what are their security implications? In what ways are contemporary strategic technology partnerships, aimed at various EDTs, similar or different from past strategic technology partnerships, such as in nuclear or cyber technology? We revisit these questions in section four.

In this article, our focus is *strategic military AI technology partnerships* specifically—that is, strategic military technology partnerships aimed at artificial intelligence technologies, comprising a wide range of approaches and architectures in machine learning.⁶

⁶ For an adjacent term, Kolodii (2020) has defined “(AI)ance,” to refer to “international cooperation in the field of AI within strategic alliances of two or more countries.”

What activities would such partnerships entail? Zoe Stanley-Lockman sketched a range of instruments in the ‘military AI cooperation toolbox’, which go beyond the pooling of resources for AI-enabled capability development, and also include “policy alignment; the testing, evaluation, validation, and verification (TEVV) pipeline; research and development (R&D), personnel exchanges; data sharing; and standardization” (Stanley-Lockman 2021, 1–2). To be sure, as Stanley-Lockman notes in this typology, such instruments need not only be implemented through multilateral alliances and partnerships, but could also be established within international defense science and technology (S&T) cooperation agreements, or the global military S&T network (id pg. 7). However, within this article, we will focus specifically on the strategic partnership angle. This is because, while states have not yet found harmonization for these procedures or standards regarding military AI, emerging AI strategic partnerships open the door for this kind of facilitation and cooperation, with important consequences for future practice.

What goals would such partnerships’ activities focus on? Whilst we do not seek to provide an exhaustive list, we could consider these to include collaborations with several different purposes; (1) facilitate the development of military AI systems amongst allies; (2) ensure the interoperability of one state’s military AI systems with the systems and practices of allies’ militaries, in order to facilitate the sale of these systems and foreign market share of companies; (3) agree on shared practices to guide the multinational operational context; (4) share live data for training and improving and updating military AI systems, and/or testing trialed systems. In more advanced cases, this could even include data fusion for integrated attack systems—cases where AI proves the ‘strategic enabler’ of what Soare has described as a “fully integrated digital and information military ecosystem comprising cyber, cloud and edge computing, and command, control and communications (C3)” (Soare 2020, 2).

A number of factors contribute to why we are seeing an increasing number of partnerships devoted to deep technological collaboration and cooperation on military AI. Geopolitical pressures have advanced technological development as a top priority in evolving strategic competition between the US, China, EU and Russia. The strategic advantage of AI on the battlefield, and the increasing potential of utilizing this technology, has propelled AI as a top security and defense priority. This clearly matters practically; yet moreover, it also matters analytically.

C. How do strategic military AI partnerships matter?

It practically matters to explore these emerging partnerships at an early stage, firstly, because their course may directly affect the range of military uses of AI and their timelines of deployment. Simultaneously, the constellation of military AI partnerships that emerges, and their characteristics, will in turn affect other AI governance mechanisms developed in parallel within the global governance architecture. This can be for numerous reasons.

In the first place, for some states, strategic partnerships *may substitute for more multilateral or normative governance initiatives*; in doing so they can strengthen and embed other norms in strategic agreements, or result in the establishment or facilitation of certain standard practices or norms around military AI development, deployment, and operation which, as argued by Ingvid Bode, can have important normative effects (Bode and Huelss 2018). Such state practice could render previously existing international regimes ‘jurisprudential space junk’ (Crootof 2019), or supersede explicit legal approaches to military AI, potentially resulting in practices that, for instance, theoretically ensure human-in-the-loop configurations, but in practice result in ‘meaningless human control’ (Bode and Watts 2021).

In the second place, military AI partnerships are valuable to explore analytically since they could *involve different dynamics and raise distinct issues* from the strategic partnerships of the past. This is because, along with key commonalities, military AI partnerships can have new features that are distinct from those involved in strategic partnerships in the past. This is because these types of collaborations will not, or need not primarily, involve the seconding of human experts, the sharing of weapon blueprints or arms transfers, or the sharing of intelligence analysis. Rather, they are likely to involve deeper shared collaborations around the sharing of larger volumes of data and trained models, including potentially sensitive data.

The sharing of large quantities of sensitive AI training data imposes its own barriers to burden-sharing and interoperability, especially within multinational military operations (Lin-Greenberg 2020). These constraints are different from those encountered in other strategic (technology partnership) contexts, as they:

1. involves the sharing of **data that is more sensitive** or even secret (compared to the data shared on other, conventional, or civilian data sharing platforms);
2. requires adequate, potentially continuous data sharing, which mandates close **interoperability in the underlying infrastructures**;

3. raises questions of **transparency and/or explainability** around military AI systems and their decisions, in ways that are not present for e.g. the sharing of conventional (human-curated) intelligence data;
4. involves **distinct expertise or human resource needs**, in terms of constrained machine learning talent, which is not always yet available or present within existing military-industrial production chains or operational theaters;
5. risks **different and at times novel failure modes** at both the tactical and strategic level of collaboration, ranging from accidental flaws in datasets, to risks of deliberate data poisoning attacks (Lohn 2021; Whyte 2020), to risks around certain AI systems' susceptibility to 'adversarial input' and spoofing attacks (Rudner and Toner 2021b).

Finally, an examination of strategic partnerships around military AI matters for scholarship on international security, sketching the contours of what may prove to be the next chapter in strategic technology partnerships. It is a valuable test of theories around strategic partnerships, and moreover provides key insights to policymakers—about how AI and its uses will be governed and managed internationally; about the changes in the role and activities of militaries; and about the potentially changing center of gravity for the governance of military AI, and for military technology more generally.

3. Examining military AI strategic partnerships: four cases

We have chosen four cases⁷ to analyze as high-profile examples of strategic partnerships to illustrate that strategic partnerships can exist along a spectrum from formal agreements to loose collaborative endeavors. Either way, each case signals a strategic political alignment and commitment to technological cooperation for future security. Importantly, these cases are not exhaustive, simply illustrative. There are many other examples of strategic partnerships for promoting military AI that are not included here.⁸ And certainly, more will emerge with time.

We begin with two examples of strategic partnerships through established agreements. First we examine the US-led AI Partnership for Defense (PfD) as a vital

⁷ A fifth possible example, the Five Eyes Partnership, is an interesting case, but is excluded from this analysis, because they are an intelligence sharing partnership (which technology cooperation as one facet), rather than a partnership based on technology cooperation and development broadly.

⁸ See for example the established strategic partnership between Singapore and the United States (Kaye 2022).

technology partnership which carries important implications for long-standing military alliances, especially NATO. Secondly, we look at the newly established AUKUS partnership between the US, UK and Australia, which includes emerging technology within its remit, especially AI. Next, we examine two examples of loose cooperation as a strategic partnership: here, we examine China-Russia AI cooperation as well as Trans-Atlantic AI cooperation, in particular between the US and the UK.⁹ For each of these, we aim to briefly (1) review recent developments and partnership strategic aims; (2) explore mechanisms of technology cooperation; (3) and outline strategic and analytical implications.

A. Partnership for Defense (PfD) - A New Kind of Strategic Partnership?

The AI Partnership for Defense (PfD) is a US-led initiative to promote collaboration for defense AI among “like-minded” allies and partners. Currently, there are 16 nations participating in the AI PfD: Australia, Canada, Denmark, Estonia, France, Finland, Germany, Israel, Japan, the Republic of Korea, Norway, the Netherlands, Singapore, Sweden, the United Kingdom, and the United States (JAIC 2021).

The partnership aims to “create new frameworks and tools for data sharing, cooperative development, and strengthened interoperability,” among the contributing partners (Ibid). For now, the PfD does *not* facilitate co-development for AI systems, but rather aims for partners to be “AI ready” to enhance later interoperability and system integration (Eversden 2021).¹⁰ The PfD represents a technology partnership established for the sole purpose of military AI cooperation - unlike partnerships in pre-existing institutions that include technology cooperation under a more general, strategic goal (like NATO and the Five-Eyes cooperation).

Instead, the PfD seeks to promote shared interests and identify problems among members, share best practices, facilitate policy alignment particularly on AI ethics standards, create frameworks for further cooperation, and coordinate strategic messaging on military and defense AI (Joint Artificial Intelligence Center 2020; Stanley-Lockman 2021).

The PfD is a model for technological partnership that will likely grow and replicate. It is a relatively low-cost style partnership that will facilitate coordination and harmonization on

⁹ Notably, in this article we focus less on the transatlantic military AI cooperation between the US and the EU, because this remains remarkably more incipient.

¹⁰ Quoting Stephanie Culberson, head of international AI policy at the JAIC DOD.

key issues related to military AI, but does not require a deep commitment to co-development or data sharing.

However, there remain uncertainties surrounding the future of the PfD. Much of the information and substance of the meetings of the PfD are not available to the public, so there is minimal information to distill substantive conclusions. To be sure, this does not mean it is an unproductive partnership – rather, the nature of the partnership prioritizes practice sharing as opposed to generating a particular product. Nevertheless, there are some salient themes we can identify about the PfD.

A first theme concerns *characteristic hurdles* that military AI partnerships continue to face. For instance, there are technical, political, and legal barriers for PfD partners that will likely hinder goals of enhanced interoperability or even lower threshold harmonization. For example, legal requirements for data sharing and management vary among the partners, and it is unclear how the partnership plans to work around this issue with sensitive defense data. Additionally, there are hurdles of uneven capabilities among partners. Largely, the contributing partners are all technologically advanced nations, but there are pronounced differences in the stages of development and political disparities regarding certain applications of military AI, such as outright weaponization.

Another key challenging theme that emerges around the PfD, concerns its *interaction with other institutions*, and specifically NATO. Undoubtedly, NATO has instruments and competencies that can positively guide Allies toward a responsible AI governance architecture (Stanley-Lockman and Trabucco 2022). Indeed, in recent years, NATO has, amongst others, established strategic initiatives on AI Horizon Scanning as well as a new Data and Artificial Intelligence Review Board (DARB), which has begun development on AI certification standards (NATO 2022; 2023; Demarest 2022).

Nevertheless, the PfD potentially offers an alternate avenue for AI problem-solving and cooperation outside the established NATO procedures or standards. This may put NATO in a difficult position as it establishes itself within the AI governance framework and prioritizing EDT's in future strategies (NATO 2021). But the PfD offers a global reach and partnerships for development that is unencumbered by preexisting conditions, trans-Atlantic borders, and bureaucratic red tape. So far, nine NATO allies are also PfD partners, though that number may eventually increase.

Conversely, a third key hurdle is that the PfD may *risk either redundancy with* collaborative actions taken within legacy institutions, or *direct competition* with activities

within those organizations. For instance, Stanley-Lockman has suggested there are many fruitful avenues for collaboration within extant institutions, noting the opportunity:

to harvest data from existing cooperative activities—such as exercises, training, and of course, operations— for AI experimentation. Within NATO and amongst the Five Eyes, the regularity of these activities makes them a natural starting point for fuller exploitation of the data. In fact, select multinational exercises incorporating dozens of unmanned systems have already set this groundwork by testing interoperability and developing tactics, techniques, and procedures. (Stanley-Lockman 2021, 32).

Given these three hurdles—(1) technical, political, and legal barriers for PfD partners that may hinder interoperability; (2) interactions with existing institutions that may erode those institutions’ activities; and the reverse (3) interactions with existing institutions’ AI initiatives that may lead to redundancy of the PfD programs—there is not yet unambiguous evidence regarding the PfD can serve as the next chapter in military AI cooperation.

It remains true that despite the availability and established cooperation through legacy institutions, states have still sought to establish AI strategic partnerships outside pre-existing security partnerships. Based on the mechanisms that the PfD is using for cooperation, the PfD serves as an important signal for global partnerships, particularly in the Pacific region, to expand the support system for AI cooperation and collaboration and likely expand American presence for AI defense integration. Additionally, the partnership may represent an alternative for the US to operate outside the NATO establishment. This will, likely, put more pressure on NATO to emerge as a leader for military AI and responsible adoption within the international security environment.¹¹ Whether such frictions can be averted or well managed will remain a key challenge for the PfD and its members.

B. AUKUS - an AI Partnership?

On September 15th, 2021, Australia, the US, and the United Kingdom (UK) announced the creation of a trilateral security agreement called AUKUS. The aims of the agreement are threefold: (1) deepen information and technology sharing; (2) strengthen the integration of

¹¹ This issue is rapidly progressing for NATO. There continue to be big developments in NATO integration of EDTs. One major step for AI is the creation of the Data and Artificial Intelligence Review Board (DARB). The board intends to be a forum for Allies to discuss responsible development of military AI and operationalize the six principles of responsible use (NATO 2022).

security and defense-related science, technology, industrial bases, and supply chains; (3) deepen cooperation on several security and defense capabilities (Prime Minister's Office 2021).

There are numerous components to the agreement, which will be discussed, but the bulk of the arrangement centered around Australia's program to create nuclear-powered submarines with the bulk of the technology coming from the US. This aspect was certainly the most publicly visible—as the agreement received significant media attention for the political snub the agreement had toward France, which had a preceding contract to supply Australia with submarines (Quinn 2021; BBC 2021). However, what was under-examined in the resulting uproar is that the AUKUS agreement includes broader technological cooperation beyond the development of nuclear submarines, also called Pillar II, which includes the development of other emerging technologies including AI. The parties to the agreement also pledge to share critical information and data regarding military AI development and quantum technologies, among others. AUKUS Pillar II is, in many ways, a critical piece of the agreement, especially as experts predict China to be ahead of the US and its partners in 19 of the 23 technologies relevant to Pillar II (Christianson, Monaghan, and Cooke 2023; Hanson and Cave 2021).

From its establishment, there are notable features of the AUKUS agreement that are distinct from other forms of technology partnerships - especially like the previously discussed PfD. However, similar to the PfD, AUKUS also operates outside long-standing legacy institutions. The AUKUS agreement takes development and technology cooperation to a far greater-degree than any legacy institution.

Additionally, at the time of announcement, there were raised eyebrows regarding the *final membership selection*: some observers raised the omission of Japan from AUKUS as significant for the Pacific region (Chansoria 2021). Despite Japan's omission, the partnership received calls for an express, deep cooperation with Japan. In November 2021, former Prime Minister of Japan, Shinzo Abe, urged Japanese ruling party leaders to maintain a close cooperation with AUKUS (Takenaka 2021). The partnership is largely considered a direct response to Chinese militarization in the region, and some in Tokyo advocate for closer ties to US presence in the region as Japan bolsters its own defense posture.

Since the initial announcement of the partnership, and more information became publicly available, it is clear that AUKUS has made significant progress in a short amount of time. The partnership focuses on clear and concrete cooperation mechanisms, including equipment and technology development and sharing, with clear deliverables, capabilities

alignment, established working groups, joint investment strategies, highly-trained worker exchange, bolstering national industry, and influencing global standards (Jackett 2022; Christianson, Monaghan, and Cooke 2023).

Unlike the PfD, the AUKUS agreement presents a deep commitment to shared technology and bolstering capabilities in an unparalleled way. AUKUS goes further than most technology partnerships toward joint-development to an impressive scale and speed. The geostrategic implications of the partnerships are clear -- rooting joint development of nuclear submarines and other emerging technology development, especially AI, is important for regional deterrence of China and enhances US (or Western) influence in the Pacific (Ibid). Historically, technology partnerships tend to fizzle out at the stage of joint-development and instead resort to national policies and interests. But AUKUS departs from this trend with important regional implications. In this sense, AUKUS is a unique strategic partnership with much more formal deliverables and joint-development than the other strategic partnerships discussed here.

C. China-Russia collaboration - a “multifaceted strategic partnership”

While most scholarly focus has been on the rising competitive US-Chinese relation around AI,¹² recent years have also seen a focus on the growing strategic relations between China and Russia (Kolodii 2020; Bendett and Kania 2019; Konaev et al. 2021). While China has historically long adhered to a ‘non-alliance policy’, aiming to avoid entrapment in unnecessary wars, this ‘alliance allergy’ does not extend to strategic partnerships (Resnick and Sworn 2023). Indeed, in light of Russian military operations in Ukraine, and the subsequent political isolation on a global scale, China emerged firmly declaring Russia its “strategic partner” and will continue to speak on a global scale with a “united voice” (Reuters 2022). A trend the Western world observers with extreme caution, if not outright alarm (Konaev et al. 2021).

The origin of this modern relationship has been linked to Xi Jinping’s 2015 state visit to Moscow, which resulted in a range of new agreements on cooperation in areas such as the digital economy (Xinhua 2016). In the years since, this collaboration has grown notably. Bendett and Kania highlight five areas in which Sino-Russian scientific and technological cooperation in AI has been expanded over the last five years: (1) dialogues and exchanges,

¹² ADD SOURCE; But for one more ambiguous take, see also Zhu and Long, “How Will Artificial Intelligence Impact Sino-US Relations?”

(2) joint investment funds, (3) the development of industrial science and technology (S&T) parks, (4) joint science competitions, and (5) expanded academic cooperation (Bendett and Kania 2019, 7). It is notable however that these types of collaborations do not yet appear to include the production of inter-operable systems. Though, recent events may change this.

Chinese-Russian cooperation mechanisms differ from Western partnerships examined in this article. China and Russia, instead, focus on sharing public surveillance and digital oversight mechanisms, significant joint-investment in technology start-ups, including AI. As of 2021, Chinese AI investments into Russia tend to focus on internet services, telecommunications, robotics and facial recognition (Konaev et al. 2021, 2). Russian AI investment in China focuses on e-commerce, autonomous driving technology, and medical services and robotics (Ibid). While these capabilities are not directly under the military AI heading, it is not difficult to imagine how these capabilities can be utilized in a military context. And finally, the creation of a Sino-Russian big data headquarters based in Harbin China (Kolodii 2020). To be sure, there is little information available regarding the

What are the drivers of this Sino-Russian strategic partnership? A key background factor is growing alignment on a vision of ‘a values-neutral global order that advantages authoritarians’ (Kendall-Taylor, Shullman, and McCormick 2020). Chinese efforts to incorporate Russia (and other allies) into the infrastructure of a Chinese technological ecosystem can both help China (and Russia) evade incorporation into a Western-dominated technological systems and help China sell Chinese products abroad (Sinkkonen and Lassila 2022). A second, more acute driver of cooperation on military technology is the perception of sustained US pressure toward both states, which has created a common cause between them, as well as an urgent strategic need for both to develop capabilities that can counter or offset growing US capabilities around EDTs (Kendall-Taylor, Shullman, and McCormick 2020). In the third place, this need to collaborate to develop these technologies is underscored by the fact that US policies towards both countries, including sanctions, export controls, and tariffs, makes both more reliant on one another as (trading) partners in the high-tech space (Edmonds et al. 2021). As noted by Kendall-Taylor, Shullman and McCormick (2020), “China has relied on imports from Russia to bolster its air combat, air defense, anti-ship, and anti-submarine capabilities. Russia also has extensive operational military experience, most recently in Ukraine and Syria, while China’s military is largely untested”.

Nonetheless, it appears that most of the focus of this partnership is still on underlying research funding and S&T exchanges, and much of this does not yet extend to the broader ‘military AI cooperation toolbox’, including TEVV procedures or testing. Indeed, it is also

unclear to what extent this program involves any preparation of data infrastructures to support coalitional or AI-enabled operations. While both countries have sought to expand the sharing of big data through the Sino-Russian Big Data Headquarters Base Project in Harbin, China (Bendett and Kania 2019, 10),¹³ this is primarily aimed at B2B digital economy cooperation, rather than military uses, and such approaches do not yet extend further, according to publicly available information.

To be sure, there may be further challenges on the horizon, including persistent concerns around copying or theft of intellectual property—although it is suggested that for the foreseeable future, such concerns “are a constraint but might not impede joint initiatives, given the potential for mutual benefit nonetheless” (Bendett and Kania 2019, 10). Nonetheless, this appears to be a remarkably high-ambition, high-stakes strategic partnership, one the fruits of which may be seen in coming years.

D. Transatlantic US-UK collaboration: continuing the special relationship

Trans-Atlantic cooperation around military AI has had a different trajectory than the China-Russia model of strategic partnerships.¹⁴ Interestingly, transatlantic ties on military AI are highly variable. For instance, there has been a notable divergence between the US and the European Union (EU), as a bloc, in their respective policy approaches to military AI (Feijóo et al. 2020), and indeed between their overarching visions for the role of AI in society (H. Roberts et al. 2021, 16–18). It is therefore remarkable that there is essentially no significant military AI strategic partnership to speak of between the US and EU, despite increasing calls emphasizing the need for greater transatlantic cooperation between the EU and the US regarding global AI governance (Lawrence and Cordey 2020). However, direct joint US-EU defense-related AI efforts remain rare. This may be because of a lack of planning or clarity of approach amongst European states (Franke 2020); a preference for increasing ‘strategic

¹³ Note, there is uncertainty over the current status of this project, as little public news has been available in recent years.

¹⁴ In fact, US military AI cooperation with the European Union as a bloc has been largely absent from the strategic partnerships landscape. Looking to recent developments, May 2023 statements of technology cooperation were very superficial with limited cooperation mechanisms – especially for the security and defense space. For instance, in the May 2023 U.S.-EU Joint Statement of the Trade and Technology Council (The White House 2023), military applications are referenced only twice. One reason for this could be the preference of European partners to engage on technology cooperation through legacy institutions -- or new partnerships like the PfD as a national member. But more research on this is warranted to better understand the lack of a strategic partnership.

autonomy’ (Olsen 2022) or, simply, a European preference to cooperate through legacy institutions or on a national level through interactions like the AI PfD. Given this incipient state, it is clear that there remain significant barriers to transatlantic defense cooperation (Soare 2020; Can and Kaplan 2020).

Importantly, that is not to say there are no European proposals. For example, the European Defence Agency (EDA) proposed its members create a repository or ‘data lake’ of less sensitive but anonymous military operational data from vehicles and air platforms, with the hope that by “giving research and technology organizations, SMEs [small-medium sized enterprises] and large industry access to it, these players could devise new AI solutions such as platform-specific smart software” (EDA 2020). Ulrike Franke has likewise noted four problematic trends, in discussing transatlantic estrangement, European digital autonomy efforts, different views on China; and, potentially, the complex relations introduced by Brexit, which has put the United Kingdom (UK), a potential defense partner, outside of the EU bloc (Franke 2021). Soare notes a range of deeper trends, such as longstanding US accusations of European free-riding on defense spending, along with “European ‘techno-Gaullism’, defence market protectionism, [and] Europe’s concerns over long-term US commitment to NATO” (Soare 2020, 4).

As such, it is clear that at present most activity around military AI cooperation along the trans-atlantic divide occurs between the UK and the US – partners with a historical special relationship. The UK published their National AI Strategy in September 2021, and articulated its Defence AI Strategy in 2022 (HM Government 2021, 47; Ministry of Defence 2022). In the latter, the UK identified ‘Key AI Partnerships’, clarifying the British relationship with the US “is our top priority for AI R&D and capability collaboration, building on our close and long-standing relationship on other technologies and capabilities” (Ibid, pg 41). The UK and US have been steadfast on collaborative R&D, facilitating academic and private industry joint ventures to foster a healthy S&T environment – even calling for the British Defence AI Centre to work alongside the Department of Defense’s Chief Digital and AI Office (CDAO) (Ibid). The intended goal is “to explore opportunities for enhanced ‘service to service’ collaboration on shared capability challenges where AI promises game-changing advantage” (Ibid).

There have been growing and increasingly concrete efforts to realize this collaboration. In September 2020, the US and UK signed the Declaration on Cooperation in Artificial Intelligence Research and Development, which created bilateral dialogue to (1) streamline existing bilateral science and technology cooperation (expanding on the 2017

Declaration of Cooperation on Science and Technology); (2) recommend priorities for future cooperation in the area of AI research and development; (3) coordinate planning and programming for experts in academia and relevant actors in the private sector; (4) promote research and development in AI embedded with democratic ethics and values (U.S. Department of State 2020; Kolodii 2020, 42–43).

In 2021, President Biden and former-Prime Minister Johnson launched a ‘landmark technology partnership’ – simply called “the Partnership” – to promote strategic cooperation to protect their technology advantage. This included addressing cross-border data flows, collaboration on semi-conductors (especially supply chain vulnerabilities), and joint concerns for AI risks and ethics (U.S. Department of Commerce 2022).

Ultimately, the British-American partnership on military AI is a promising and deep commitment between governments and industries toward promoting responsible and ethical AI. Both countries focus on embedding democratic and liberal values to counter-balance much of the potential from partnerships that may not share the same interests - particularly the Chinese-Russian strategic partnership and areas of potential.

4. Reflections: mechanisms of cooperation, future directions

The map of strategic collaboration on military AI remains complex and in flux. Each of the cases above illustrate a type and form of AI strategic partnership. They vary in their origins, goals, degree of commitment and cooperation, and pathways by which they operationalize cooperation. Nonetheless, each fulfills the core criteria for a ‘strategic partnerships’. As such, they illustrate the range and spectrum of strategic partnerships; the diversity of cooperation mechanisms (Table 1), and their distinct implications for geopolitics and AI.

Table 1: Cooperation Mechanisms Characterizing the different Partnerships

AI Partnership for Defense • Best Practices sharing • Problem identification • Knowledge sharing • Strategic messaging	AUKUS • Equipment and technology sharing • Prioritization on deliverables (for regional deterrence) • Capabilities alignment • Created working groups • Detailed investment strategies • Influencing global standards • High-trained workers exchange
China-Russia • Sharing on public surveillance & digital oversight • Chinese companies biggest investors in	US – UK AI collaboration • Declaration(s) of Cooperation • Streamline existing cooperation strategies • Plan programs for joint academic

• Russian AI • Joint investment in start-ups • Establishing a Sino-Russian big data HQ in Harbin, China	• ventures/private industry ventures • Promote R&D
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A few patterns suggest themselves. One thing that is clear is that military AI strategic partnerships must negotiate questions of formality, breadth (in their membership) and depth (in terms of the cooperative mechanisms committed to).

For instance, the AI Partnership for Defense (PfD) is a defined, formal partnership that extends membership beyond US partners in legacy institutions and incorporates technologically like-minded states with shared political and strategic interests. Certain technologically advanced states, such as China and Russia, are clearly not included in the PfD due to these geopolitical alignments. But there are still other notable membership omissions. For example, it is significant that India has also not joined the PfD, especially given that India and the United States collaborate in other institutional mechanisms that include technological cooperation.

In terms of depth, one might expect at first sight that the clearly defined membership of the PfD, and its potential for greatly enhanced interoperability and data sharing and cooperation in future military operations suggests this partnership may be a crucial mechanism shaping military AI cooperation in the future. To be sure, as military partners and Allies continue to integrate or function in coalition settings, especially in the Pacific region, partnerships like the PfD could help overcome the initial hurdles technology can pose for coalition integration and harmonization. Nevertheless, looking at the concrete mechanisms of cooperation suggest the PfD is for now a largely symbolic forum with surprisingly modest influence on the development of military AI systems. To be sure, outright joint development was not the purpose of the PfD. The PfD, instead, operates as an outlet for members to share best practices toward responsible military AI systems and seek shared knowledge or support for tackling pressing challenges unique to each members' approach to military AI, national values, or national defense industries. As such, the PfD constitutes a tailored (in breadth), and low-risk (in depth) engagement for its members, relying on cooperation mechanisms that are standard fare for many other Western strategic partnerships.

AUKUS, by contrast, is the most visibly formal strategic partnership and departs from most technology strategic partnerships of the past by focusing heavily on joint-development and is accelerating at a significant pace. In April 2022, AUKUS released a joint statement detailing progress at six months, including descriptions of specific projects already underway

(Ministry of Defence 2023). For AI projects, the AUKUS nations are developing undersea autonomous vehicles with trials and experiments planned as quickly as 2023, demonstrating the depth and pace of AUKUS joint-development. All of its other cooperation mechanisms (such as advanced workforce-exchange, joint investment, and influencing global standards) are much more concrete and advanced cooperation mechanisms than those evident in other strategic partnerships, especially those involving the US.

As such, AUKUS is a high-cost, high-reward partnership that prioritizes deterrence in the Pacific region. Indeed, when the partnership was initially announced, it was a significant political snub to France due to the contract with Australia for nuclear submarines - another example of the political cost associated with committing to this partnership (Quinn 2021). As such, these types of (non-bilateral but exclusive-membership in breadth; significant in depth) strategic partnerships are likely to be rare, but could be potent drivers at advancing military AI research generally, and at deploying many autonomous systems in particular.

The Russia-China partnership is a step in a different direction. As a self-proclaimed “strategic partnership”, their informal technological collaboration represents a political alignment of convenience on the global stage, in contrast to Western alliances. Both China and Russia are vocal about the strategic importance of EDTs, and AI in particular, which is why it is interesting, even peculiar, that a more formal agreement resembling the PfD or AUKUS does not yet exist between these states (and others). This is especially true because, as the surveyed cooperation mechanisms suggest, there is very concrete cooperation that, in some cases, extends deeper than do those actions by Western partnerships. For example, the announced creation of big data headquarters in Harbin, China, is a major step to operationalize cooperation – a move that has not yet been seen from Western counterparts.

Importantly, the Russian-Chinese relationship was recently tested during the Russian-Ukraine conflict, and China’s muted response to such aggression is a strong signal for the future of Chinese-Russian relations. However, while China did not condemn Russia directly, and rather directly condemned NATO’s eastward expansion, it also abstained during a UN Security Council vote condemning Russian military actions (Nichols and Pamuk 2022). This ambivalence is, of course, relevant for the future of Chinese-Russian political and technological cooperation. As Russia is increasingly isolated from certain technological sectors due to sanctions, including technology sanctions, Russia will likely increasingly rely on Chinese support and may seek to establish firm cooperation and collaboration avenues to maintain a technological edge in future military operations. At the same time, given its

increasing isolation, this may lead Russia to serve as a ‘junior partner’ in the strategic partnership, even more (and even more visibly) than has been the case so far.

The trans-Atlantic space is notable in particular for the absence of a strong strategic partnership between the EU bloc and the US on military AI. One reason for this might be that European partners simply prefer to foster military AI development and cooperation through legacy institutions, especially NATO. Certainly, NATO is active in exploring, monitoring, and developing EDTs that will advance the Alliance’s mission and enhance future coalition effectiveness; but the same problems remain to integrated and effective research and development programs, or the establishment of information architectures that could support coalitional operations. European states have fundamental differences regarding certain implementations, particularly weaponization, of AI, and these divergences may impede future directions.

Nevertheless, there is substantial cooperation between the US and UK toward AI innovation and development. Based on formal arrangements between both partners, the priority seems to be AI embedded with democratic values and to maintain joint technological edge. There is an emphasis on joint-development: however, much of the details of this cooperation in the defense space is not necessary publicly accessible and remains classified. Nevertheless, there is significant engagement between the two countries, a strong bent toward fostering public-private partnerships, and efforts to generate mutual academic innovation to continue to advance the technological space. Yet in other ways, we still do not observe concrete cooperation mechanisms, such as establishing a big-data headquarter, in the military space. There has been cooperation on intelligence data, especially surrounding issues like surveillance – but little information is known about defense-specific AI data sharing (Kolodii 2020).

Ultimately, strategic partnerships around military AI are permeating the international security landscape. On the one hand, some partnerships (such as the PfD and AUKUS) emerge outside traditional boundaries and legacy institutions to foster cooperation and address the pressing challenges – and potential opportunities – that AI creates for future warfare. Other partnerships (such as Russia-China and US-UK cooperation) represent bilateral forms of cooperation as part of long-standing strategic and technology engagement. . Such cases illustrate the rich and varied weave of the fabric of strategic partnerships—and the sheer diversity of ways that States can conceive of, approach, and operationalize cooperation on emerging disruptive technologies such as AI.

Conclusion

This article has aimed to provide a systemic exploration of a new and notable trend in technology and geopolitics: the growing role of strategic partnerships around the development and use of military AI technology. The emergence of strategic partnerships have significant implications for AI's future military use and play an important role in the geopolitical environment.

The four examples detailed above illustrate how military AI strategic partnerships are shaping the landscape of technological development and, potentially, military practices and partnerships in the implementation and use of AI. We reviewed the background of developments in military AI, discussed work aimed at mapping international cooperation around it, and providing a rationale for examining strategic partnerships. We then set the conceptual foundations for the debate, discussing the concept of strategic partnership, and the particular challenges facing military AI strategic partnerships. We discussed four examples of recent developments in this landscape: the AI Partnership for Defense; AUKUS; China-Russia collaboration, and broad transatlantic collaboration (in particular between the US and the UK), and before finally providing some longer-term reflections on the prospects for- and implications of these initiatives.

The emergence, development, and success of military AI strategic partnerships has a long way to go. This landscape is steadily evolving and shifting as political, legal, and ethical considerations also shape any future implementation and established practices of military AI. Nevertheless, it is important to understand the role of strategic partnerships within the emergence of military AI and how these partnerships enable, or advance, technological innovation. Strategic partnerships help shape the landscape for the emergence of military AI, and will likely be important to future military cooperation, integration, and practice.

There is still a long road ahead for understanding the implications of AI strategic partnerships, as this space is currently developing, and doing so quickly. Many of the four partnerships discussed are still in their infancy and the full scope of their implications and potential is still being written. Nevertheless, it will be a key area of development, for those hoping to study, or shape, the next chapter in the complex relation between technology, security, and international relations.

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